

Video Game AI - Decision Tree CRUD for NPC Behaviour



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Introduction :

NPCs (Non-Player Characters) make decisions in games

Actions include:

Attack

Patrol

Run

Search

Decision Trees help simulate intelligent behaviour

ROBOTIC

Problem statement:

NPCs need smart decision-making

Manual logic is complex

Need system to:

Add decisions

Update decisions

Delete decisions

Search decisions

Solution: Decision Tree + CRUD Operation



Data Structure Used:

Binary Tree

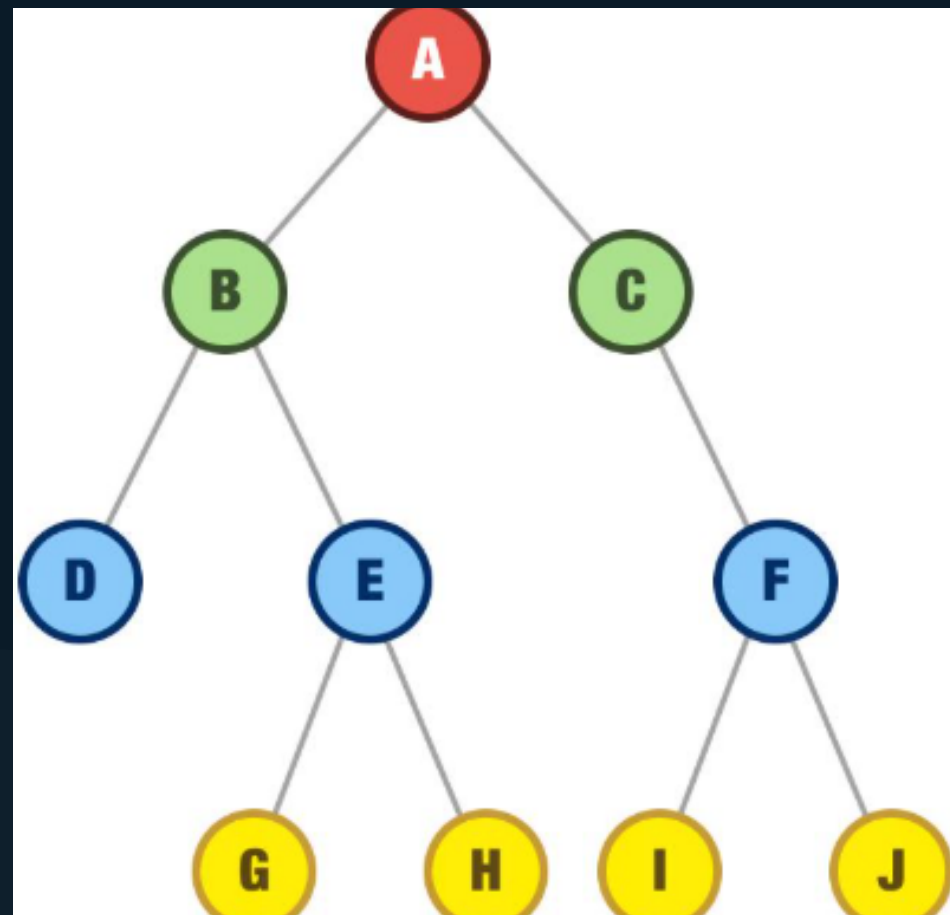
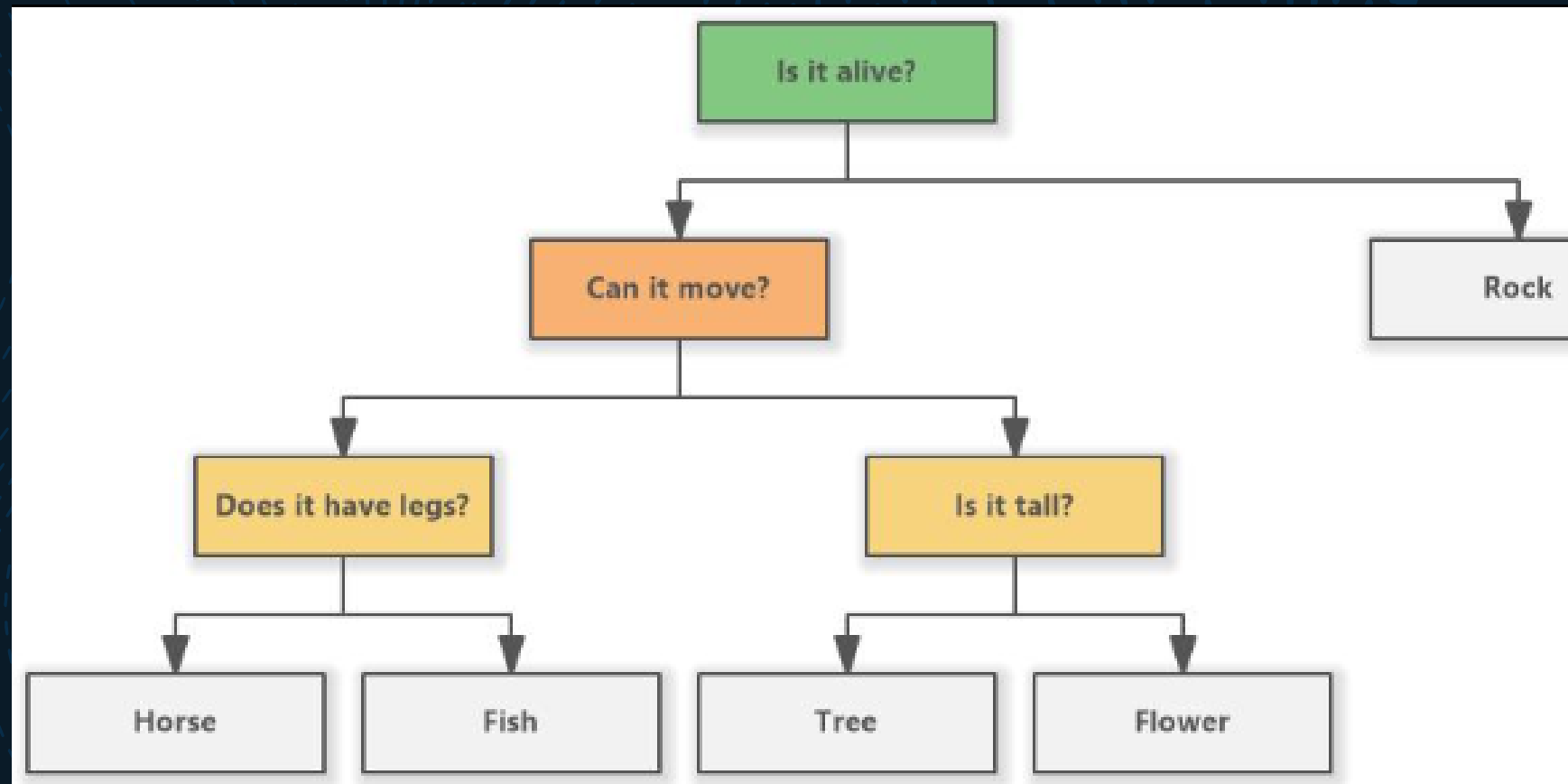
Node contains:

ID

Decision

Left child

Right child



Node Structure(Code)

```
struct Node
{
    int id;
    char decision[100];
    struct Node *left;
    struct Node *right;
};
```

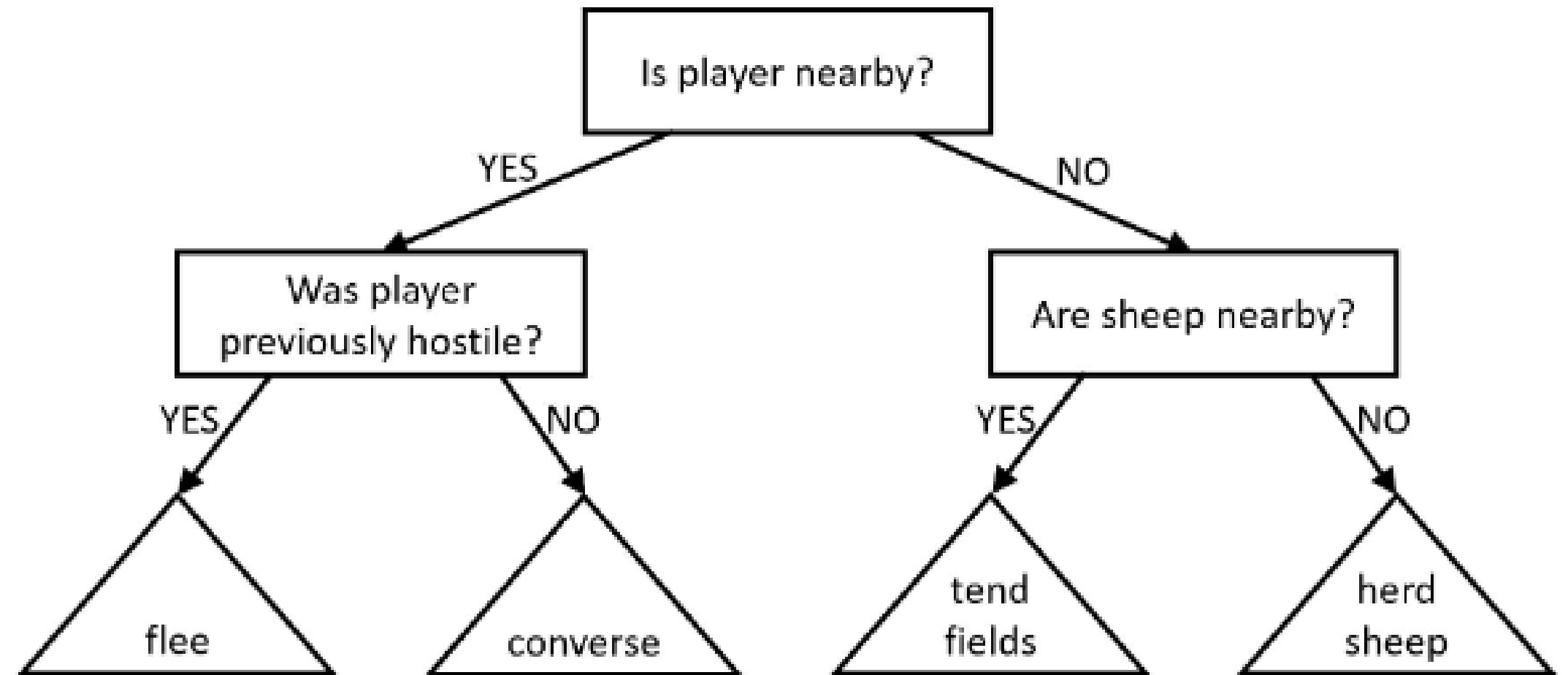
System Design

Example Decision Tree:

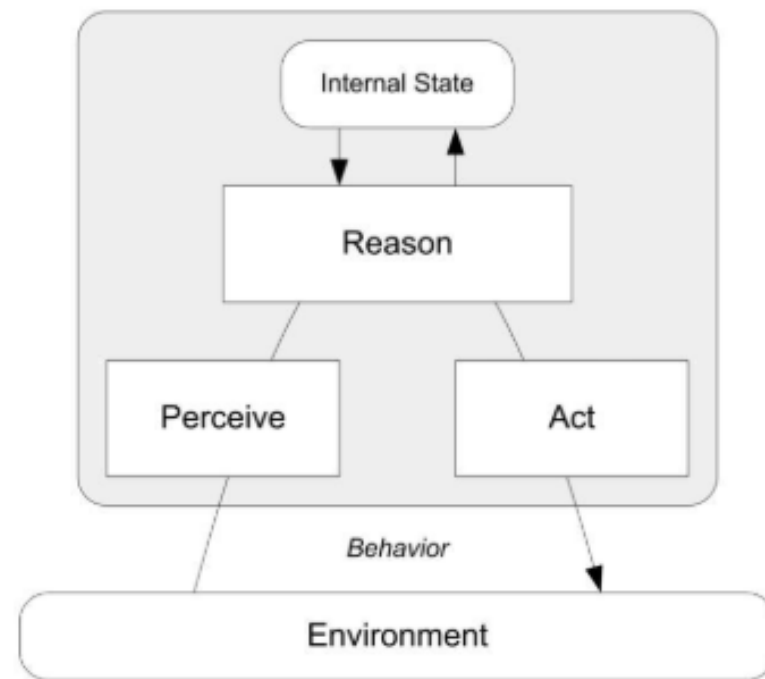
Enemy Nearby?

Yes → Attack Enemy

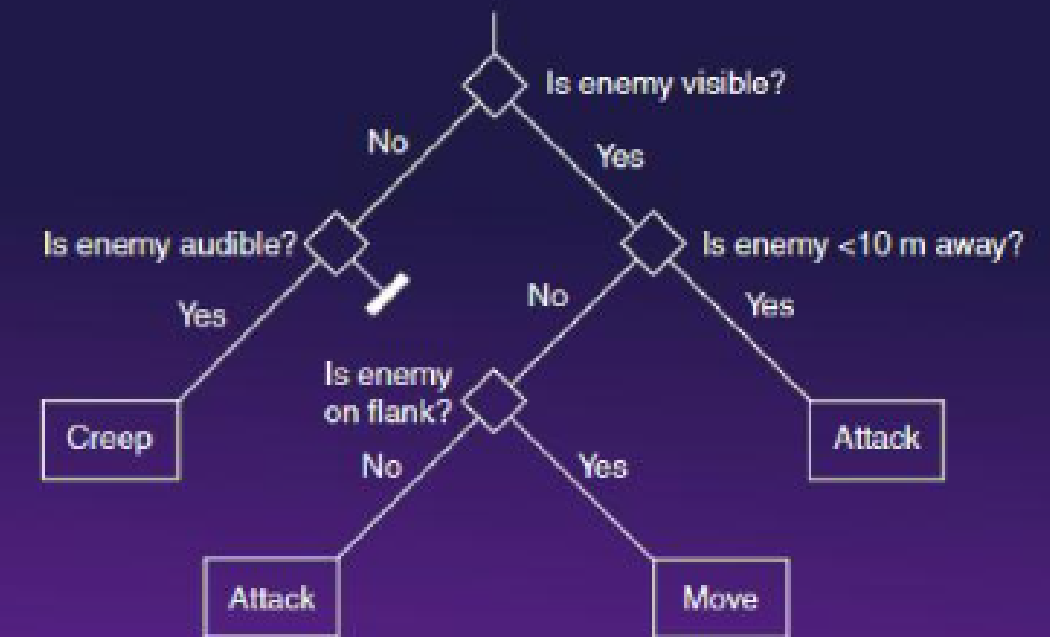
No → Patrol Area



NPC Behavior

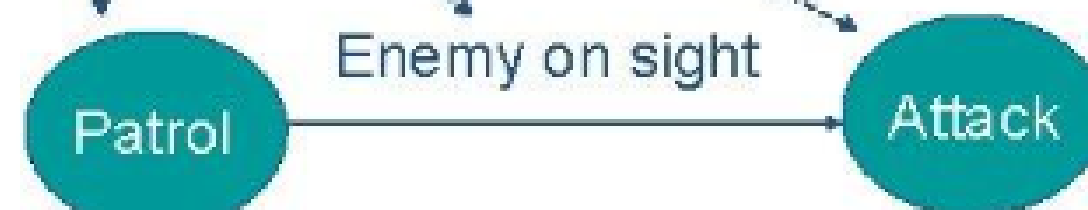


Decision Tree



NPC's Behavior is Modeled through Finite State Machines (I)

- **State**: an activity performed by an avatar
- **Event**: something that happens in the game world that makes state change



CRUD Operations

Create → Add Node

Read → Display

Tree Update → Modify

Decision Delete → Remove

Node

Algorithm (Insert)

1.Start

2.Check if tree empty

3.Create node

4.Compare IDs

5.Insert left or right



Algorithm (Search)

1. Compare ID with root
2. If equal → Found
3. If smaller → Left
4. If larger → Right Algorithm (Update & Delete)

Update:

Search node

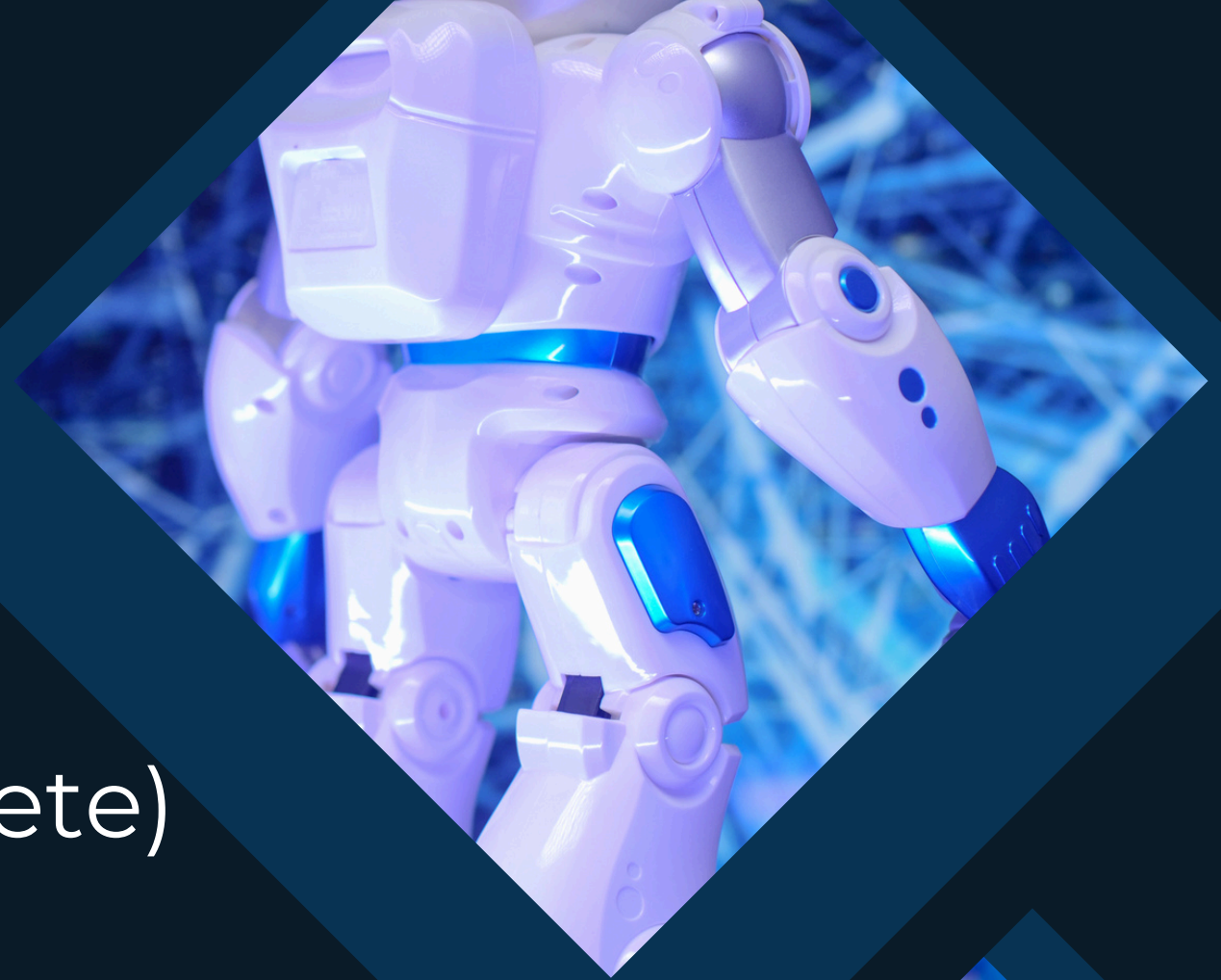
Modify decision

Delete:

Find node

Remove it

Adjust tree



Implementation Details:

C Programming Structures

malloc & free

Functions

Tree traversal

Menu Operations

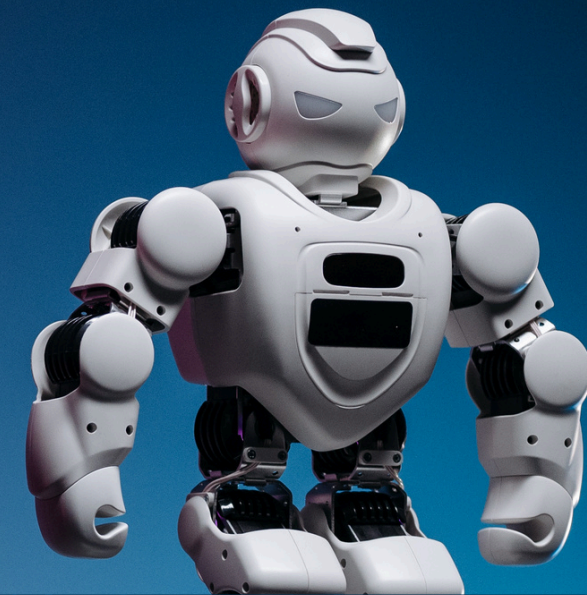
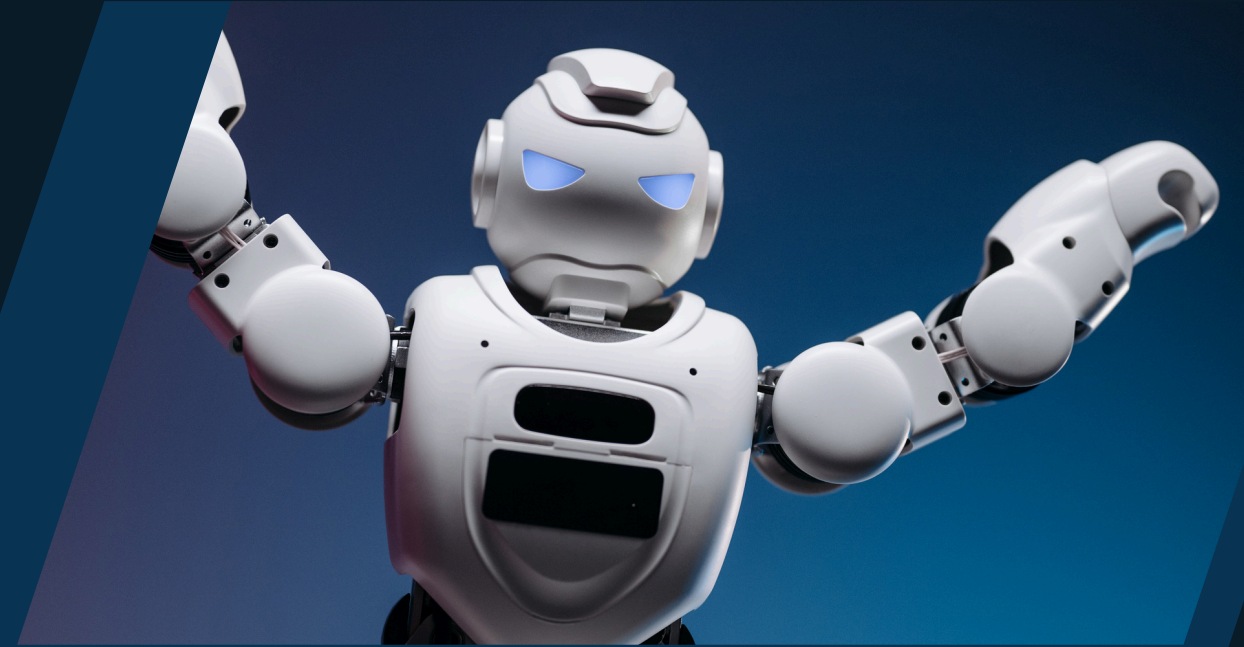
Add Node

Delete Node

Update Node

Search Node

Display Tree Exit



Limitations :

Console-based only

No graphical view

Basic AI logic

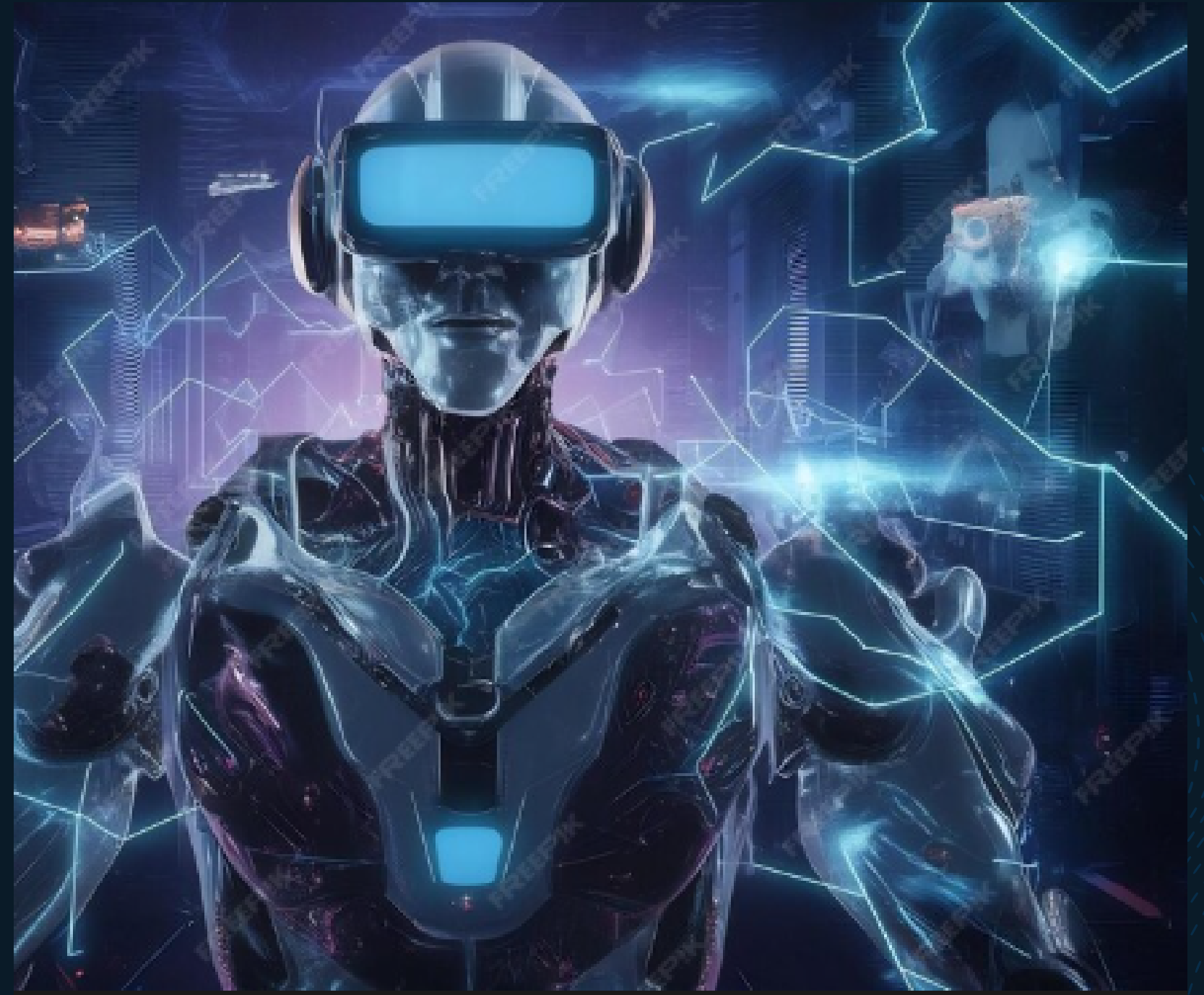
Future Enhancements

Graphical tree visualization

Game engine integration

Advanced AI logic

Multiple conditions



Conclusion :

Implemented Decision Tree

CRUD operations successful

Useful for Game AI

Learned data structures & memory

Thank you